

3.2.1 Types of Functions (Built-in vs User-defined)

بلٹ ان فنکشن پائتھن میں پہلے سے موجود ہوتے ہیں، جیسے `len()` اور `print()`.

CS XII

Chapter 3: Programming Fundamentals

Lecture 10 of 18

3.2.1 Types of Functions (Built-in vs User-defined)

LEARN IT

There are two main types of functions in Python.

a) Built-in functions are already defined by Python. You just call them. Examples you have already used:

`len()`, `range()`, `sum()`, `print()`, `input()`, `min()`, `max()`, `sorted()`, `int()`, `float()`, `type()`.

b) User-defined functions are functions **you** create with `def` for tasks that no built-in covers in one step — greeting a user, dividing with a zero check, writing a grade file.

	Built-in	User-defined
Who writes it?	The Python language	The programmer
How you start	Just call the name	First write <code>def name(...):</code>
Examples	<code>len("Ali")</code> → 3	<code>def add(a, b): return a + b</code>
Purpose	Everyday operations needed by everyone	Your program's specific jobs

WORKED EXAMPLE

Side by side. Counting letters is built-in. Combining a greeting with a name is user-defined.

```
print(len("Sindh"))

def greet_student(name):
    print("Hello,", name)

greet_student("Hassan")
greet_student("Malika")
```

OUTPUT

```
5
Hello, Hassan
Hello, Malika
```

You did not write the code inside `len`. You did write the code inside `greet_student`, then reused it for two names.

REMEMBER IT

If Python already has it (`len`, `sum`, `max`), do not rewrite it. Write a user-defined function when you are packaging **your** steps.

COMMON MISTAKE

Do not name your function `list`, `print`, `sum` or `input`. You would hide the built-in name for the rest of that file. Choose names such as `print_bill` or `sum_marks`.

اردو میں سمجھیں

بلٹ ان فنکشن پائتھن میں پہلے سے موجود ہوتے ہیں، جیسے `len()` اور `print()`۔ یوزر ڈیفائنڈ فنکشن وہ ہیں جو پروگرامر `def` سے خود بناتا ہے تاکہ مخصوص کام بار بار کیا جا سکے۔ جو کام بلٹ ان پہلے کر دیتے ہیں انہیں دوبارہ نہ لکھیں۔

CHECK YOURSELF

1. Is `range()` built-in or user-defined?
2. Which keyword creates a user-defined function?

Answers: 1. Built-in. 2. `def`.

Next lecture: 3.2.2 Functions with Return Values